



GATEWAY TECHNICAL

Supervised Learning: Regression



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1. Introduction to Supervised Learning

Supervised learning maps inputs to outputs using labeled data

Definition

Learn mapping from inputs (X) $\rightarrow$ output (y)

Regression vs. Classification

- **Regression** $\rightarrow$ continuous outputs (numbers)
- **Classification** $\rightarrow$ categorical outputs (labels)

Regression problems arise across business and scientific domains

Business examples

- Predict housing prices from property features
- Forecast sales based on advertising spend
- Estimate customer lifetime value

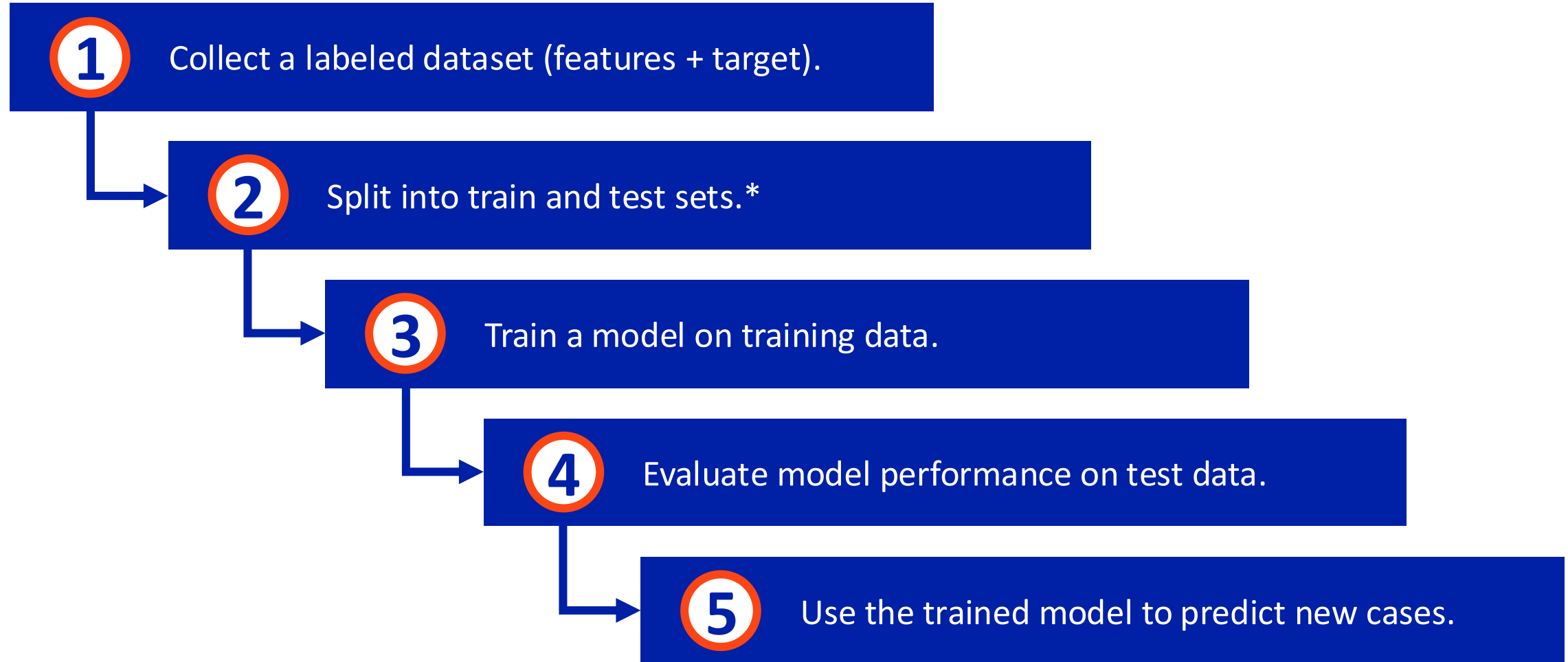
Scientific examples

- Predict blood pressure from medical measurements
- Model energy consumption from weather and usage patterns

An end-to-end ML pipeline connects problem definition to deployment

- 
- 1 **Is there a relationship?** Do the predictors explain variation in (y) beyond noise?
 - 2 **How strong is the relationship?** Effect sizes; variance explained (e.g., R^2), RSE)
 - 3 **Which predictors matter *uniquely*?** Conditional effects “holding others fixed”; avoid misattribution
 - 4 **How big are the effects (with uncertainty)?** Coefficients + **CIs/p-values**; practical vs. statistical significance
 - 5 **How accurate are predictions?** **Confidence intervals** for the mean vs. **prediction intervals** for individuals
 - 6 **Is a linear model appropriate?** Residual diagnostics; consider **polynomials/interactions** if needed
 - 7 **Are there interactions or synergies?** Do effects of one predictor depend on the level of another?

The supervised learning workflow follows a standard* process



** This is not always standard... should it be?*

Regression can be seen as function approximation

Goal

Learn the underlying relationship between inputs and outputs

Categorical variables

$$y = f(X) + \epsilon$$

- $f(X)$: Systematic relationship
- ϵ : Random error/noise

The task is to approximate f closely while generalizing to **unseen** data.



2. Linear Regression

Ordinary Least Squares regression provides a simple and interpretable baseline

Idea	Fit a straight line (or hyperplane) that minimizes squared errors.
Simple Regression Equation	$y = \underbrace{\beta_0}_{\text{Intercept}} + \underbrace{\beta_1 x}_{\text{Slope}} + \underbrace{\epsilon}_{\text{Error Term}}$
Ordinary Least Squares	OLS chooses coefficients to minimize: $SSE = \sum (y_i - \hat{y}_i)^2$
Why it matters	Baseline that is interpretable and often competitive

Population vs. sample regression lines

Population regression line

$$Y = \beta_0 + \beta_1 X + \epsilon$$

True but unknown relationship in the data generating process

Sample regression line

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

Estimated from our finite dataset

Key idea

OLS gives the **best linear unbiased estimate (BLUE)**.
Different samples → slightly different 'lines', but on average close to truth

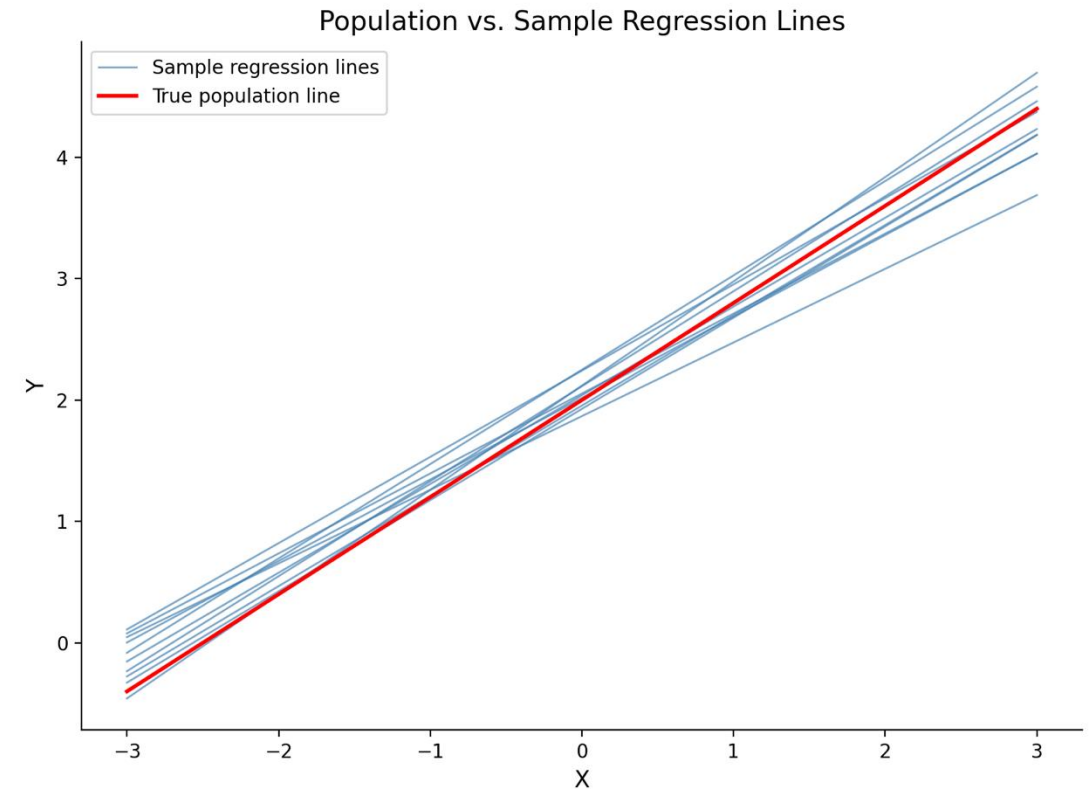


Figure: True line in red, sample regression fits in blue across repeated samples

OLS finds the line that minimizes squared errors

- Each prediction error = residual: $(e_i = y_i - \hat{y}_i)$.
- OLS chooses coefficients $(\hat{\beta}_0, \hat{\beta}_1)$ to minimize: $RSS = \sum_{i=1}^n e_i^2$

Intuition

- Imagine adjusting slope and intercept.
- Each choice gives a different RSS.
- The OLS estimates are the coordinates that give the smallest RSS.

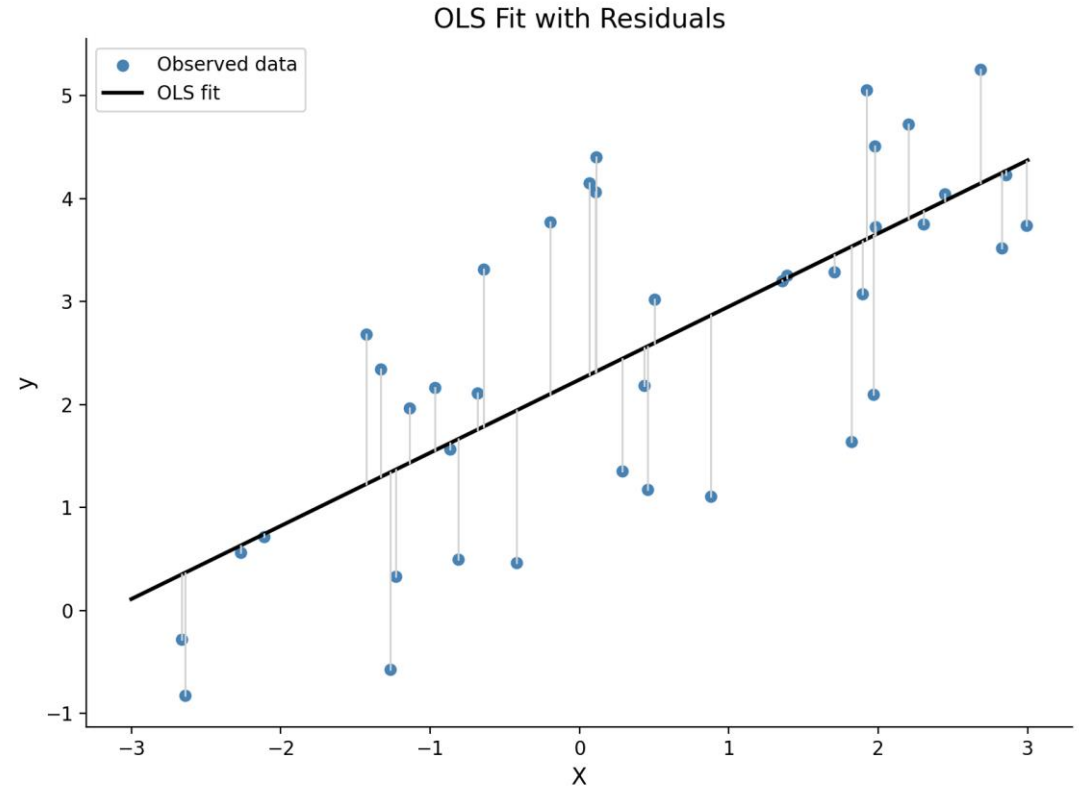


Figure: Scatterplot with residuals shown as vertical lines

Visualizing: RSS can be graphed as a **surface** over values of (β_0) and (β_1) .

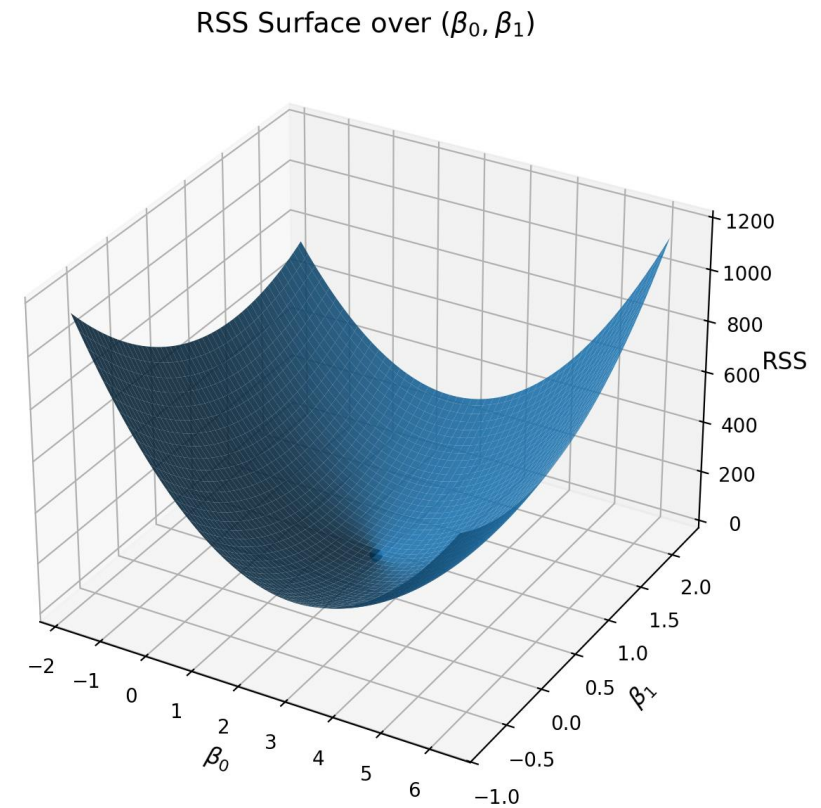
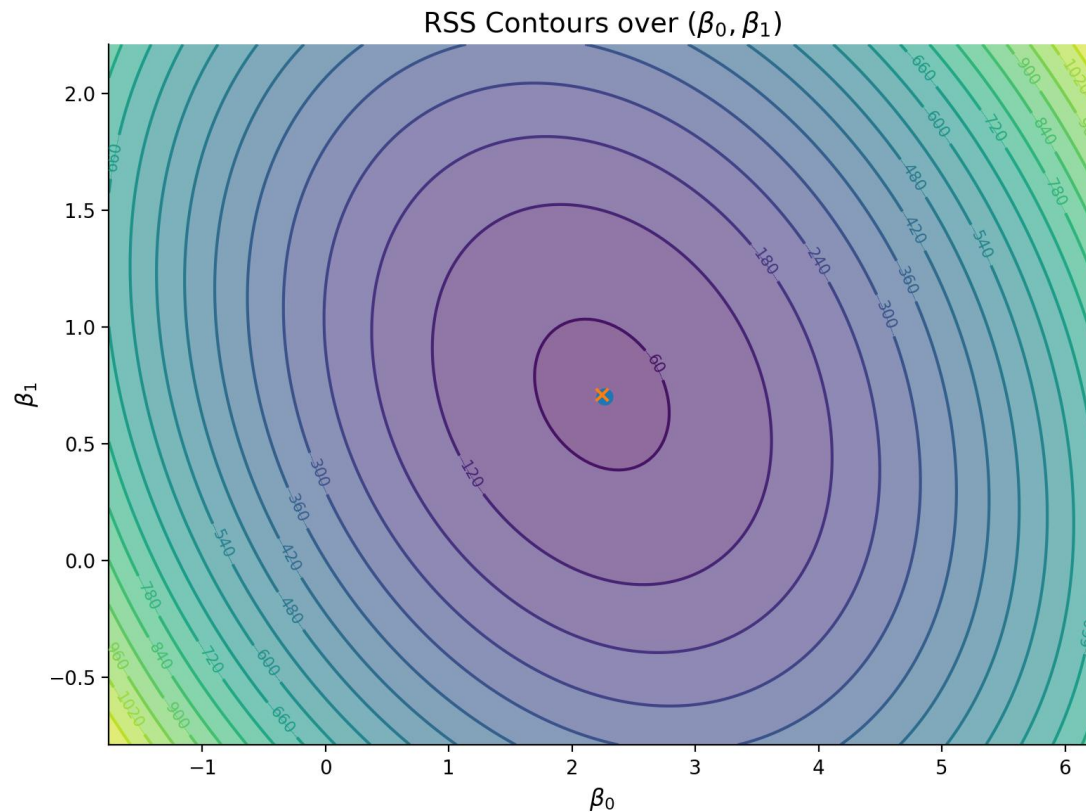


Figure: Contour plot + 3D bowl of RSS surface

Linear regression relies on several core assumptions

Linearity	Predictors relate linearly to the outcome
Homoscedasticity	Residuals have constant variance
Normality of residuals	Residuals follow a normal distribution
Independence	Observations are independent
Full rank	No perfect multicollinearity among predictors
 Note	Violations may harm inference, though predictions can still work

Coefficients and intercepts have clear interpretations



Intercept

Predicted value of y when all features are 0*

Coefficients

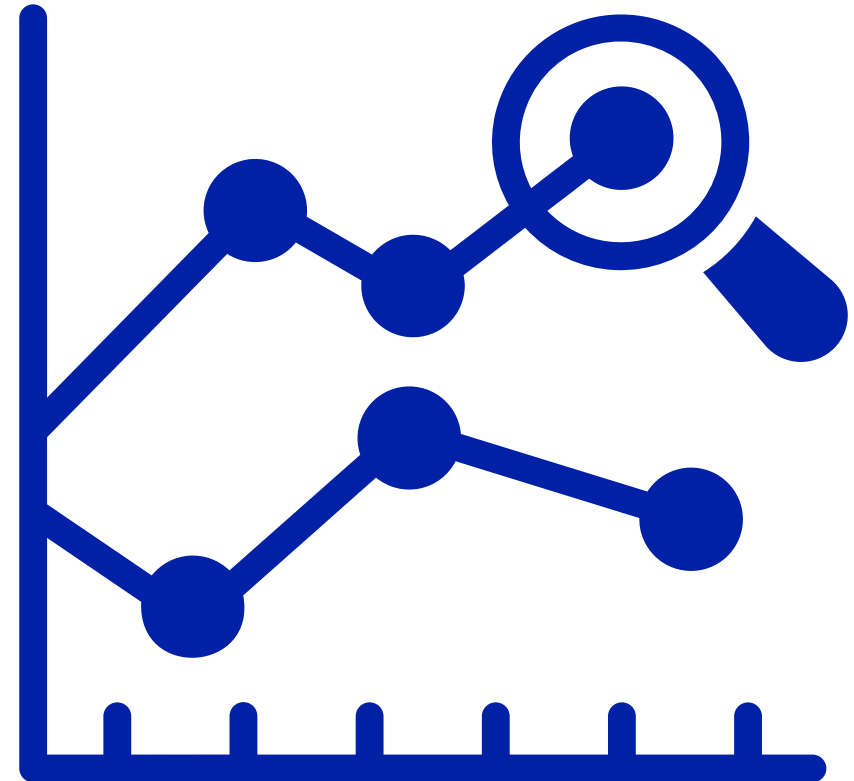
Change in y for one-unit change in x , holding others constant

Example

Coefficient on “size” = 250 → each additional square foot increases price by \$250

Regression inference asks: Does this predictor really matter?

- Coefficients are estimates, not exact truths.
- We need tools to measure **uncertainty**:
 - Standard errors
 - Confidence intervals
 - Hypothesis tests
- These tools help distinguish real effects from random noise.



Hypothesis tests check if coefficients differ from zero

Null hypothesis

$$H_0: \beta_j = 0$$

Alternative

$$H_a: \beta_j \neq 0$$

Test statistic

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

Small p-value

Strong evidence that predictor matters*

* But p values aren't everything

Confidence intervals show a plausible range for effects

95% confidence interval

$$\hat{\beta}_j \pm 2 \times SE(\hat{\beta}_j)$$

Interpretation

If we repeated the study many times, 95% of intervals would contain the true effect.

Example

$\beta(\text{size}) = 250$, $SE = 25 \rightarrow CI = [200, 300]$

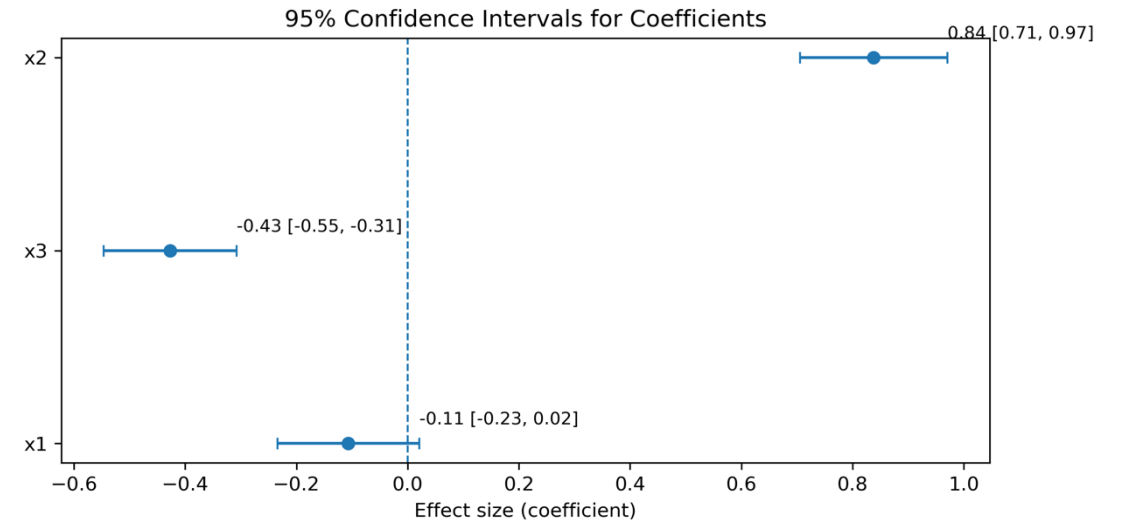


Figure: Confidence intervals show a range of possible effects



3. Regression Diagnostics

Visualization of predictions and residuals helps diagnose fit

Prediction plots

True vs. predicted values

Residual plots

Residuals vs. predictions:
Look for randomness

Histogram of residuals

Check normality assumption

Example

Funnel-shaped residual plot = heteroscedasticity

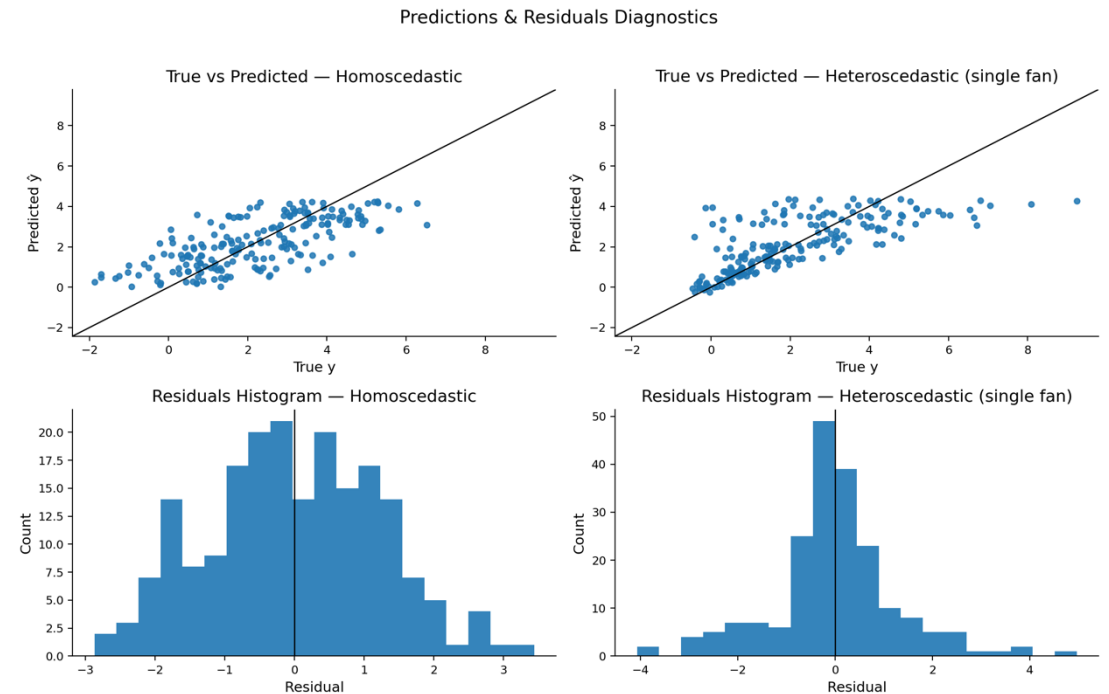


Figure: Scatter and residual plots for regression model

Diagnostics reveal hidden problems in regression models

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- Assumptions often fail in practice.
 - Without diagnostics, results can be misleading.

Key issues

- Outliers
- High leverage points
- Multicollinearity
- Heteroscedasticity or nonlinearity

Outliers and leverage points can distort results

Outliers	Unusual response values → large residuals
High leverage	Unusual predictor values → strong influence on regression line
Normality of residuals	High residual + high leverage
Dangerous combination	• Studentized residuals • Leverage statistic
Tools	No perfect multicollinearity among predictors



Check Cook's Distance as a combined influence measure; large values flag points with high leverage × large residual.

Outliers Example

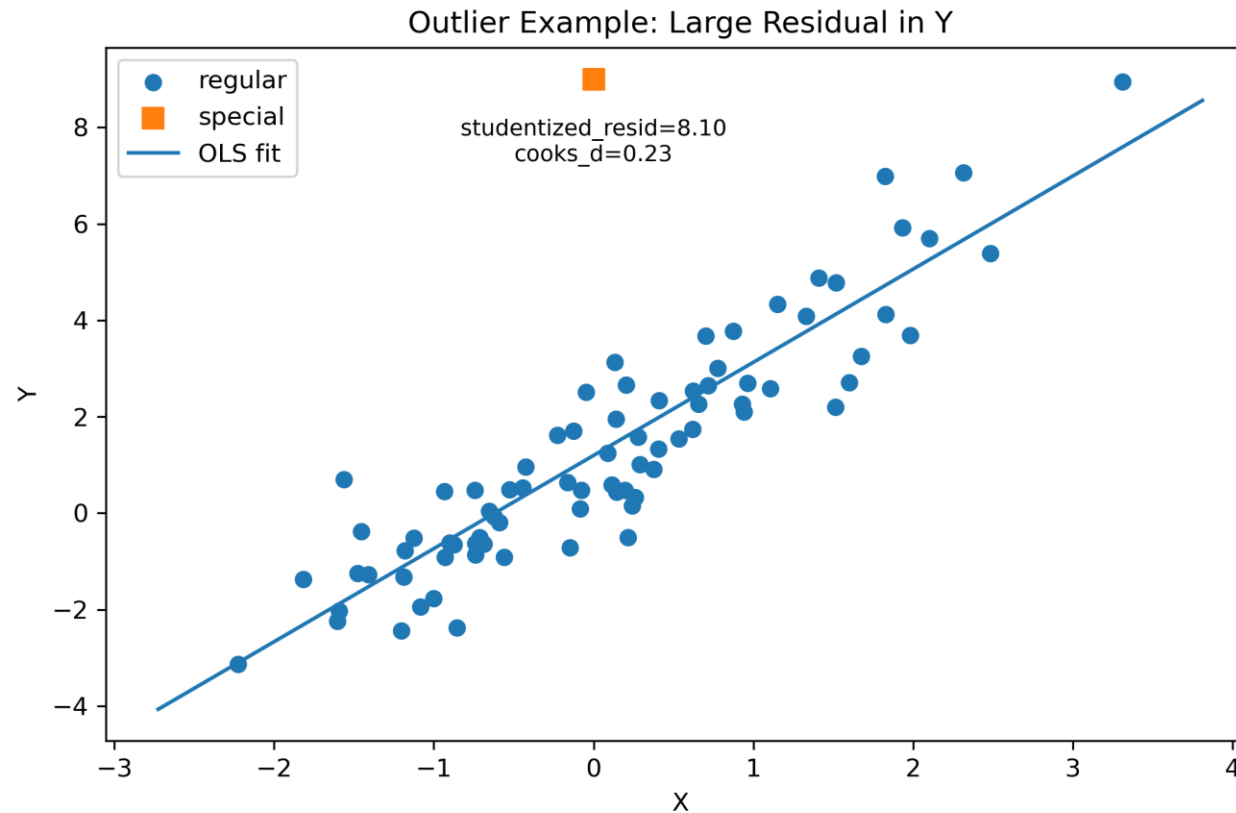


Figure: Outlier example: typical x , large vertical error.
Annotation shows external studentized residual r^* and Cook's D

High Leverage Example?

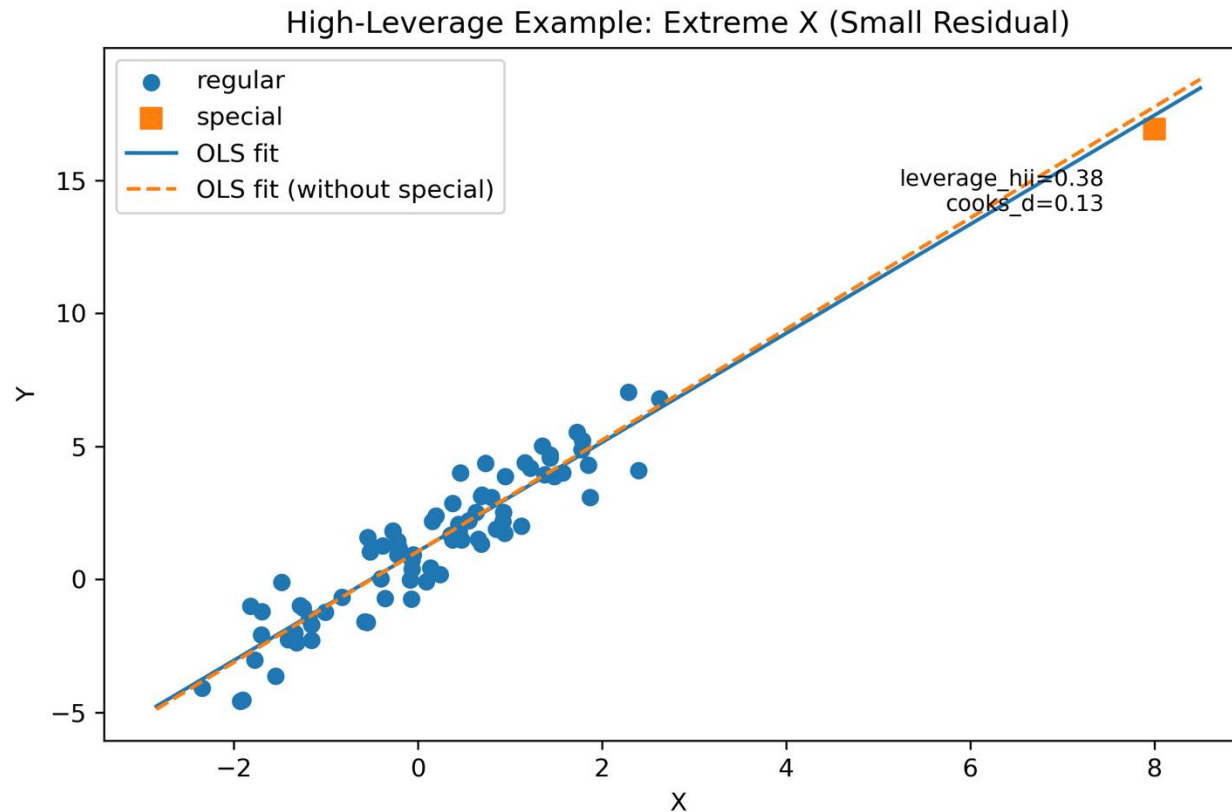


Figure: High-leverage example: extreme x , small residual. Solid line = OLS (all points); dashed line = OLS without the leverage point. Annotation shows h_{ii} and Cook's D

An outlier with high leverage is a bad combination!

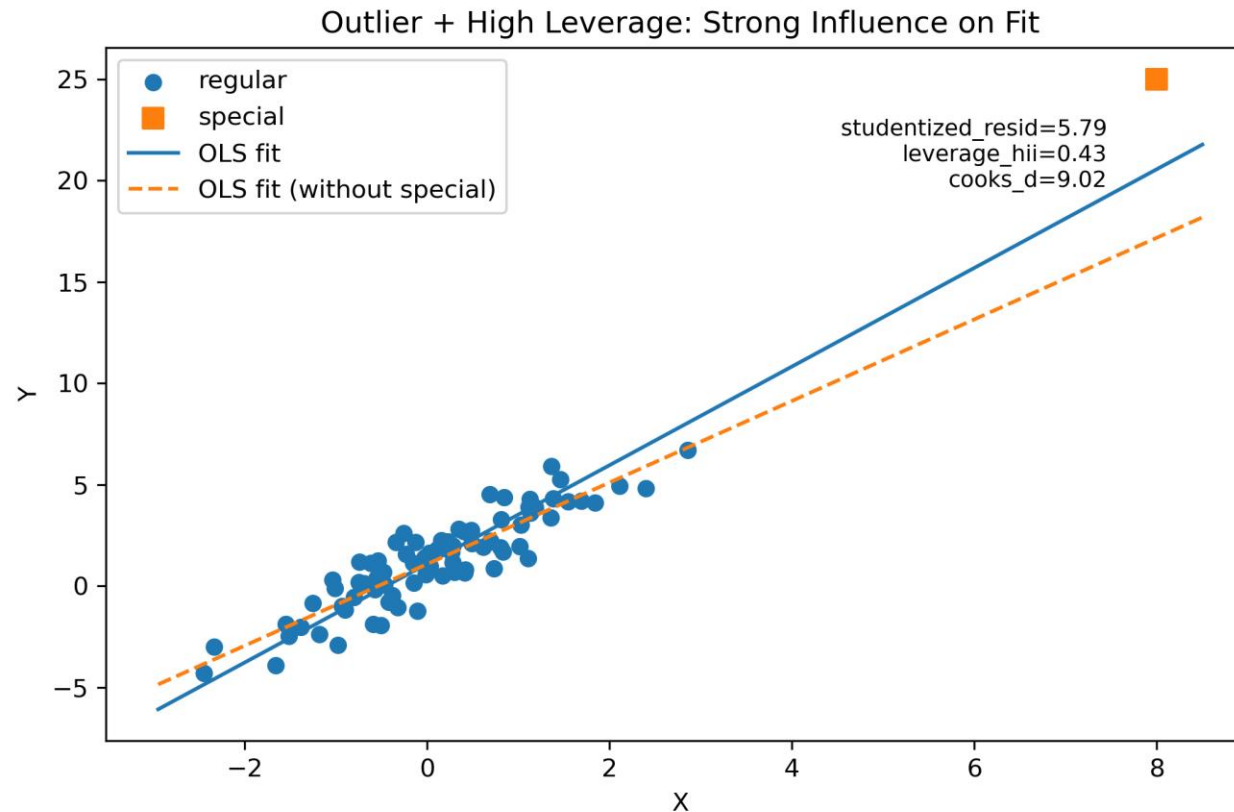


Figure: Outlier + high leverage: extreme x and large residual. Both lines shown; annotations include r^* , h_{ii} , D . Note how one point can “tilt” the line.

Multicollinearity makes coefficients unstable

- When predictors are highly correlated → hard to separate effects
- Coefficients become unstable with inflated standard errors

Variance Inflation Factor (VIF)

$$VIF(\beta_j) = \frac{1}{1 - R_{X_j|X_{-j}}^2}$$

Rule of thumb

VIF > 5 (or 10) = problematic

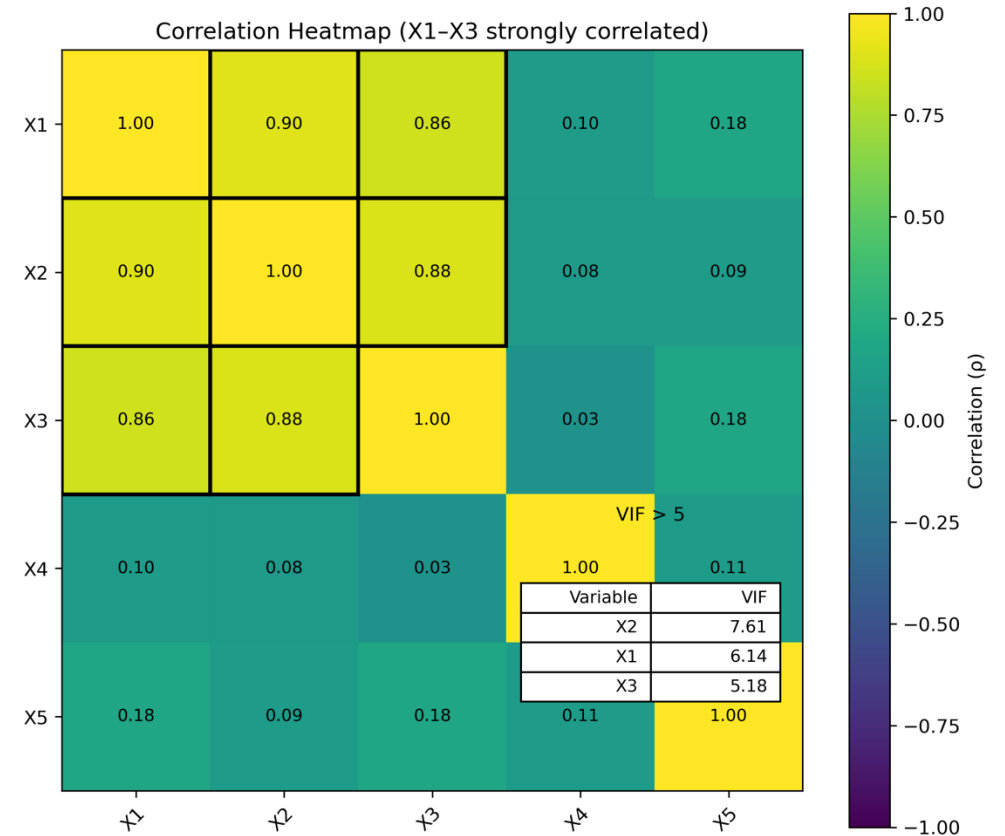


Figure: X1–X3 show high pairwise correlations, signaling potential redundancy

Residual Standard Error (RSE): Error in original units

Residual Standard Error	Estimates the average deviation of observations from the regression line
Formula	$RSE = \sqrt{\frac{RSS}{n - p - 1}}$ • RSS = residual sum of squares • (n) = sample size, (p) = number of predictors.
Expression	Expressed in the same units as (y)
Interpretation	“Typical prediction error”

Example: Interpreting RSE

Advertising model

Sales ~ TV + Radio + Newspaper

RSE =

1.69 (thousands of units)

Mean sales ~

Sales ~ TV + Radio + Newspaper

Interpretation

- Predictions are typically off by ~1,690 units.
- Error ≈ 12% of the mean → fairly good fit?

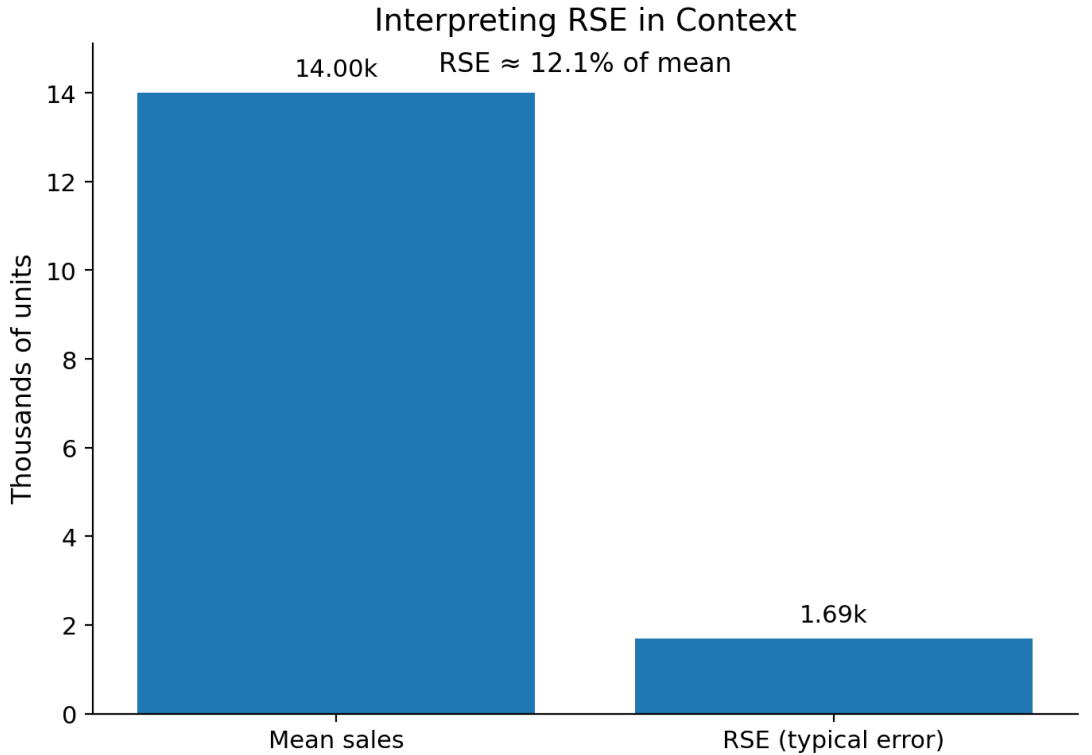


Figure: Bar chart showing mean sales vs. average RSE error

RSE vs. R^2 : Complementary measures

RSE

- Error in original units
- Easy to communicate as “average prediction error”

R^2

- Proportion of variance explained (0–1)
- Scale-free, useful for comparisons across problems

Taken together

- RSE = absolute sense of fit
- R^2 = relative sense of fit

Regression assumes errors are independent (one error shouldn't predict another)

Linear regression assumption

$$\text{Cov}(\epsilon_i, \epsilon_j) = 0 \quad \text{for } i \neq j$$

Violations common in:

- Time series data (sales by month, stock returns)
- Clustered data (observations from the same person, family, store)

If ignored → standard errors too small → false confidence.

Detecting correlated errors

Plot residuals against time/order of data

- Random scatter $\rightarrow$ independence holds
- Patterns/trends $\rightarrow$ correlation present

Example patterns

- “Runs” of positive/negative residuals
- Slow-moving cycles in residuals

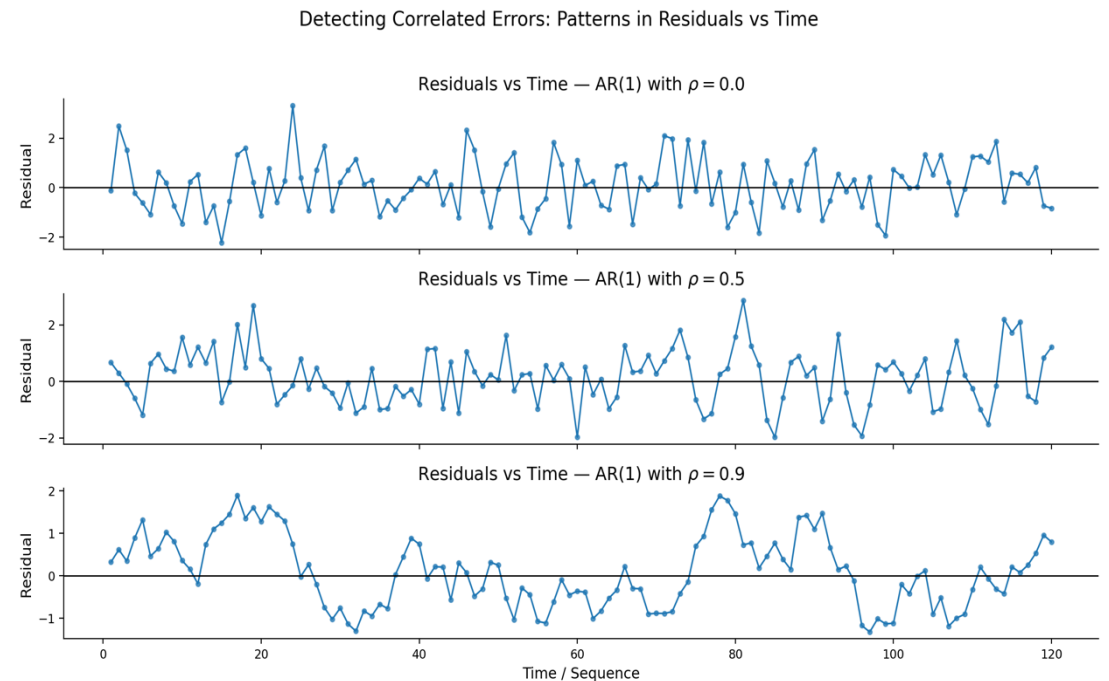


Figure: Residuals plotted against time;
top = random, bottom = clear trend

Remedies for correlated errors



Time series models

ARIMA, exponential smoothing

Robust standard errors

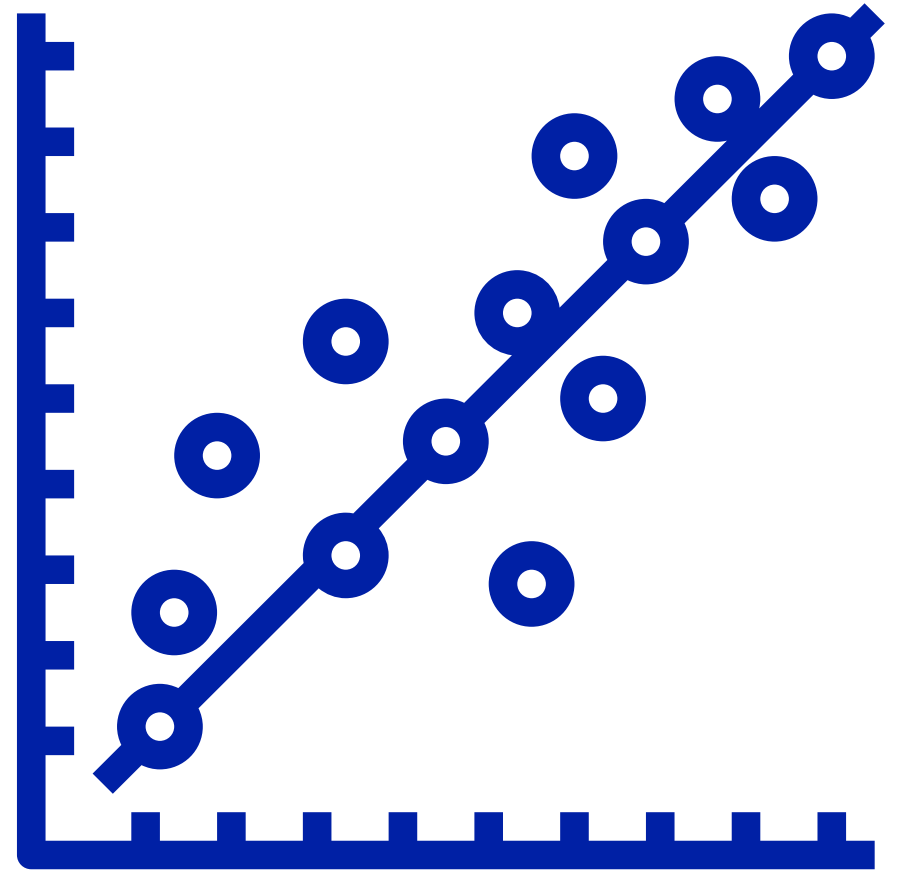
Adjust inference when clustering exists
(e.g., cluster-robust SEs)

Design changes

Collect data to minimize dependence (randomization)

Linear regression is a valuable baseline for both learning and practice

- Interpretable, fast, and intuitive
- Serves as a benchmark against more complex models
- Provides diagnostic insight into data before moving to nonlinear methods





4. Multiple Regression

Multiple regression extends linear regression to many predictors

Equation

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

- Allows us to estimate the effect of each predictor **holding others constant**
- **Answers:** Which predictors add unique explanatory power?

Separate regressions can exaggerate or misattribute effects

- Predictors often correlate with each other.
- **Example:** Advertising spend on TV and newspaper both rise with sales.
- Simple regressions may credit newspaper for an effect actually driven by TV.
- Multiple regression separates unique contributions.



Coefficients in multiple regression have a conditional meaning

- β_j : Expected change in y for a one-unit increase in x_j , **holding all other predictors fixed**.
- **Example:** TV coefficient = 0.046 $\rightarrow$ each extra \$1,000 on TV ads increases sales by ~ 46 units, holding radio and newspaper constant.
- Interpretation is always **conditional** on the other variables in the model.



Beyond individual predictors: does the model matter?

- In multiple regression, we may ask: Are **any** predictors related to the response?

Null hypothesis

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

Alternative

At least one $\beta_j \neq 0$

- Tested using the **F-statistic**

The F-statistic compares explained vs. unexplained variance

Formula

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$$

- **TSS:** Total sum of squares (total variation in (y))
- **RSS:** Residual sum of squares (unexplained variation)
- **Numerator** = variance explained per predictor
- **Denominator** = variance left per degree of freedom

Large (F)

Strong evidence that at least one predictor matters

Example: Advertising data

Fit

Sales $\sim$ TV + Radio + Newspaper

Output

- Residual standard error (RSE) = 1.69
- ($R^2 = 0.897$)
- F-statistic = 570 ($p < 0.0001$)

Conclusion

- The overall model explains much more variance than chance.
- At least one medium (TV/Radio) is strongly associated with sales.

Multiple testing caution in regression

In multiple regression with many predictors

- Running many t-tests → some small p-values appear **by chance**
- **Example:** With 100 predictors, expect ~5 to be “significant” at 5% level even if none truly matter

Consequences

- False positives (spurious significance)
- Overstated confidence in weak predictors

Solutions

- Use **overall F-test** to check if model adds value
- Prefer **cross-validation** to assess predictive usefulness

Point predictions aren't enough

Two interval types for uncertainty:

(1) Confidence interval (mean response) and (2) Prediction interval (individual)

Confidence interval (CI)

Average effect: Quantifies uncertainty around the mean response at given (X)

Formula (approx.): $\hat{y} \pm 2 \times SE(\hat{y})$

Example:

- Spending \$100k on TV ads predicts 11.2k sales
- 95% CI for mean sales = [11.0k, 11.4k]

Prediction interval (PI)

Individual outcome: Quantifies uncertainty around a single new observation

Formula: $\hat{y} \pm 2 \times SE_{\text{pred}}$

Always wider than a CI, since they include irreducible error (ϵ)

Confidence vs. prediction intervals: Side by side

Confidence interval (CI)

- “Where the mean will fall”
- **Narrower:** Used for inference on group averages

Prediction interval (PI)

- “Where an individual will fall”
- **Wider:** Used for forecasting single cases.

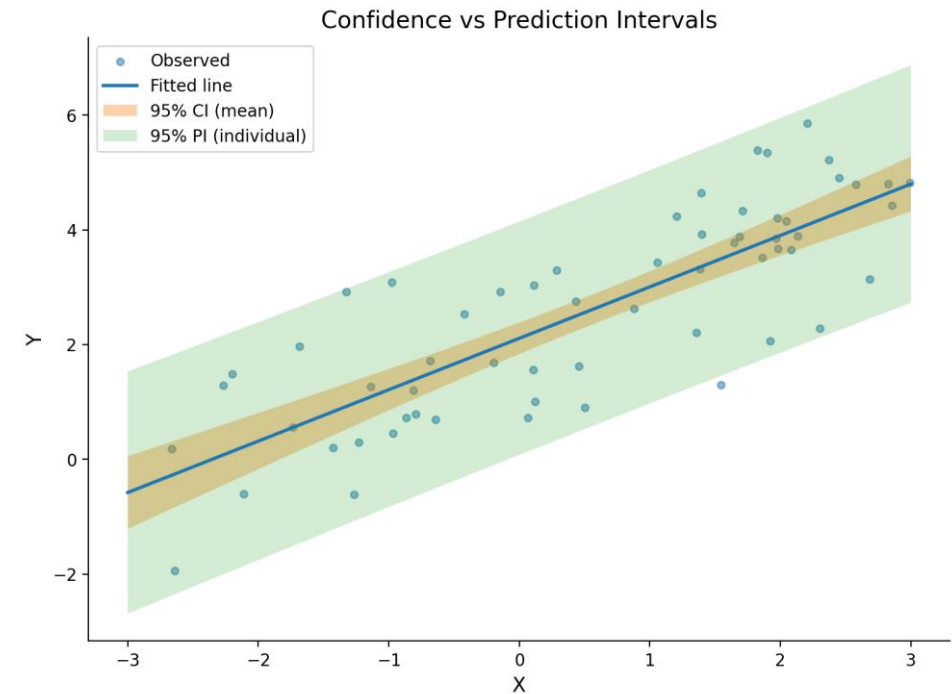


Figure: Graph with regression line, narrow CI band, wider PI band



5. Interactions & Categorical Predictors

Predictors may interact with each other

Standard linear regression assumes additive effects

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Effect of (X_1) does not depend on (X_2)

But often one predictor **modifies** the effect of another

Solution

Include an **interaction term** $(X_1 \times X_2)$

Example: interaction between two continuous predictors

Model

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2) + \epsilon$$

Interpretation

- Effect of (X_1) changes depending on (X_2) .
- **Example:** TV and Radio advertising. TV more effective when paired with Radio.
- $(\beta_3 > 0)$ indicates synergy.

Interaction Surface: TV $\times$ Radio ($\beta_3 > 0 \Rightarrow$ Synergy)

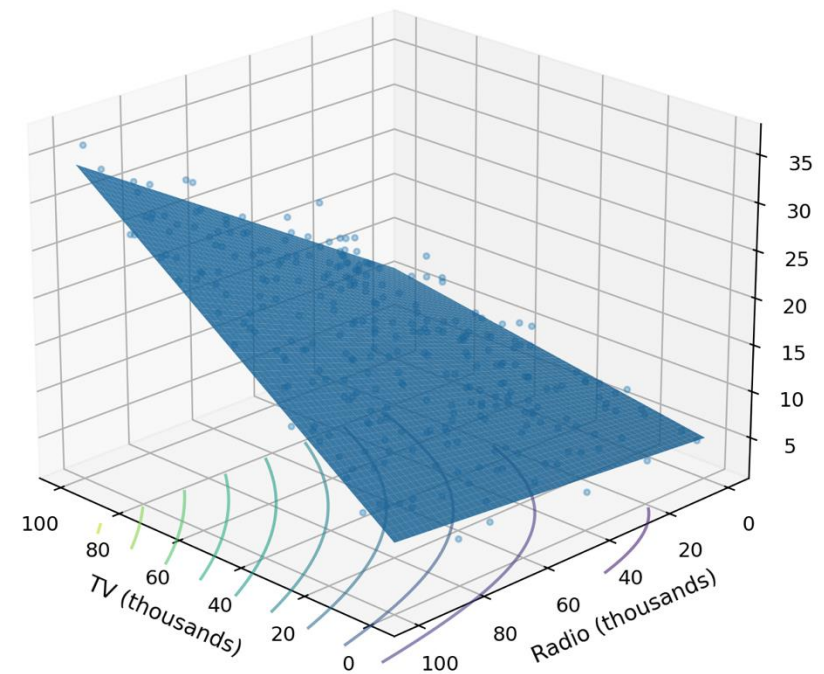


Figure: 3D plot of sales vs. TV + Radio; curved surface contour lines shows synergy

Example: interaction between category and number

Suppose predictors

- Income (numeric)
- Student status (Yes/No)

Model without interaction

Same slope, parallel lines for students and non-students.

Model with interaction

Slopes differ by group

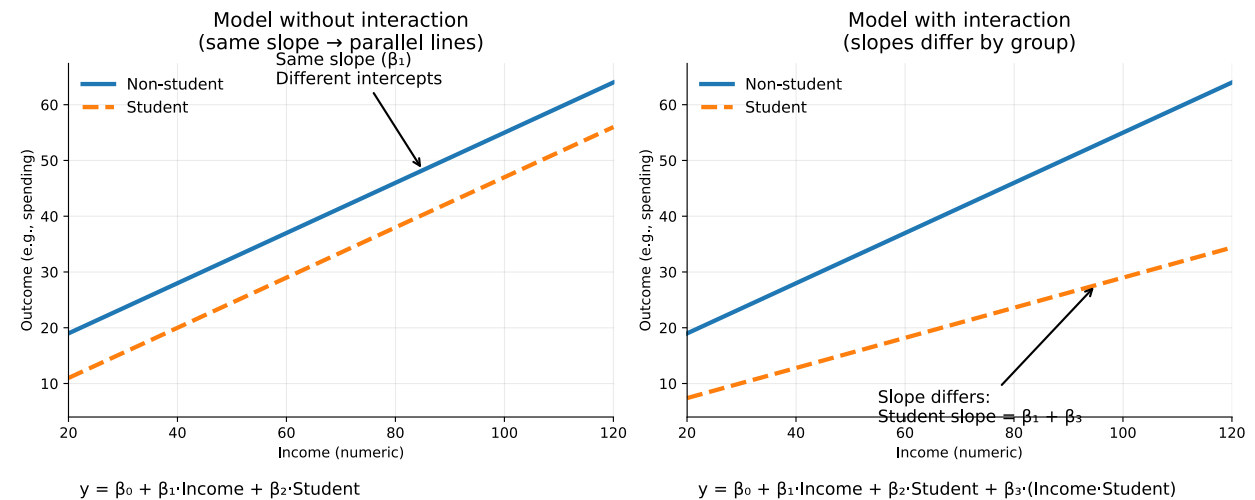


Figure: Two regression lines, parallel vs. differing slopes

The hierarchical principle



Rule of thumb

If an interaction is included, keep the main effects too.

Example

Even if (β_1) (TV) not significant, keep it if TV×Radio is included.

Reason

Dropping main effects changes the interpretation of the interaction.

Regression handles categories with dummy variables

- Many predictors are **categorical** (e.g., gender, region, product type).
- Dummy variable = indicator (0 or 1) for category membership

Example: Own a house (Yes/No)

- Own = 1, Rent = 0.
- Equation: $y = \beta_0 + \beta_1 \cdot Own + \epsilon$

Coefficients represent differences from a baseline

- **Intercept (β_0)** = predicted outcome for baseline category
- **Dummy coefficient (β_1)** = interpretable as the difference between baseline and category

Example

- **Non-owners:** Average balance = \$510
- **Owners:** Average balance = \$510 + β_1

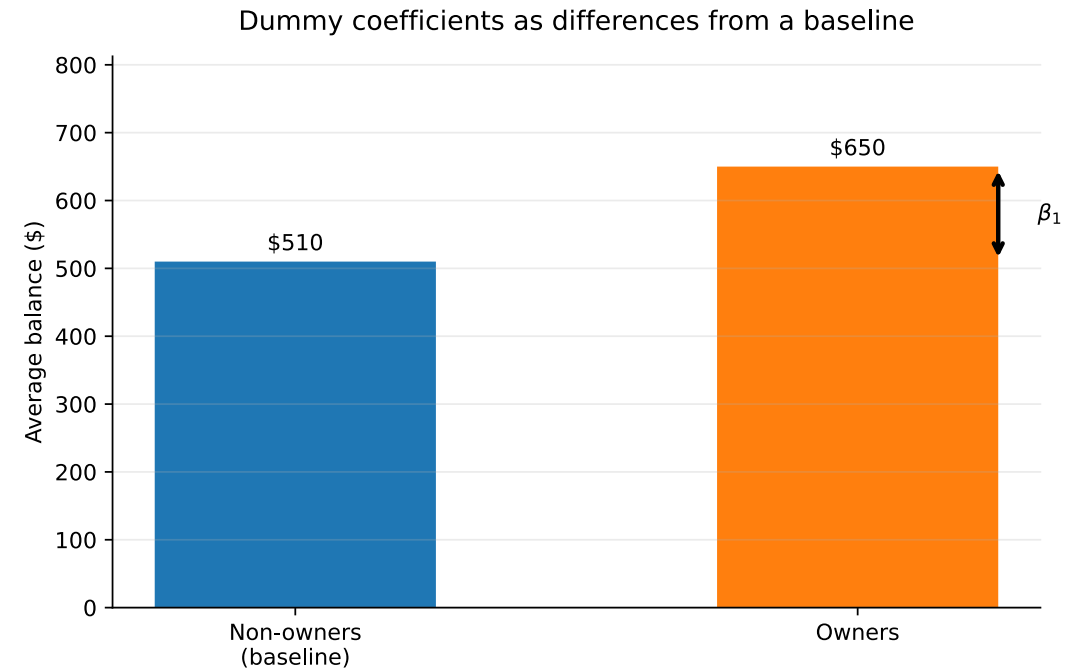


Figure: Dummy coefficient are interpretable as the difference from the baseline category

More categories → need multiple dummies

- For k categories, create $k-1$ dummy variables.

Example: Region (East, South, West)

- **East** = baseline
 - **South dummy** = 1 if South, 0 otherwise
 - **West dummy** = 1 if West, 0 otherwise
- Coefficients measure differences from the baseline.

Region	South (Dummy)	West (Dummy)
East	0	0
South	1	0
West	0	1

Figure: Table of region encoding into dummies. East is the baseline category

The dummy variable trap

- For k categories:
Use $k - 1$ **dummies**, not k .
- If all k are included $\rightarrow$ perfect multicollinearity
 - Model matrix becomes non-invertible.
 - OLS cannot estimate coefficients.

Example: Region (East, South, West)

Include only 2 dummies; baseline = East

Region	South (Dummy)	West (Dummy)	East (Dummy)
East	0	0	1
South	1	0	0
West	0	1	0

Figure: With 3 categories coded, the model can no longer estimate coef.



6. Polynomial Regression

Polynomial regression extends linear models with transformed features

- Linear regression can be extended by adding polynomial terms of predictors.

Example: Single Variable

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon$$

- This is still a **linear model in the parameters** (β) ;only the inputs are transformed.

Residual plots reveal nonlinearity

- If relationship is nonlinear, residuals show patterns.

Example

Linear model: residuals show a U-shape.
(This example is from **horsepower.csv**.)

- A straight line misses systematic structure.

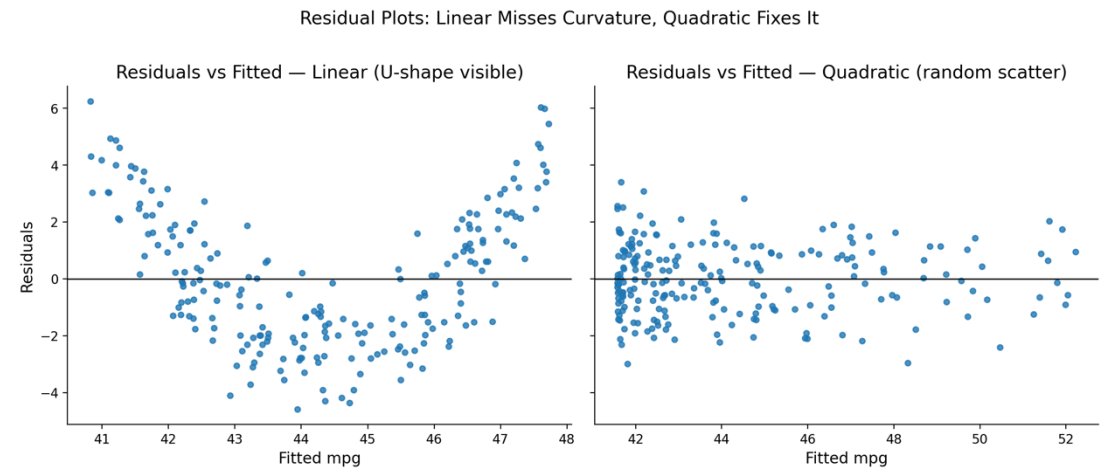


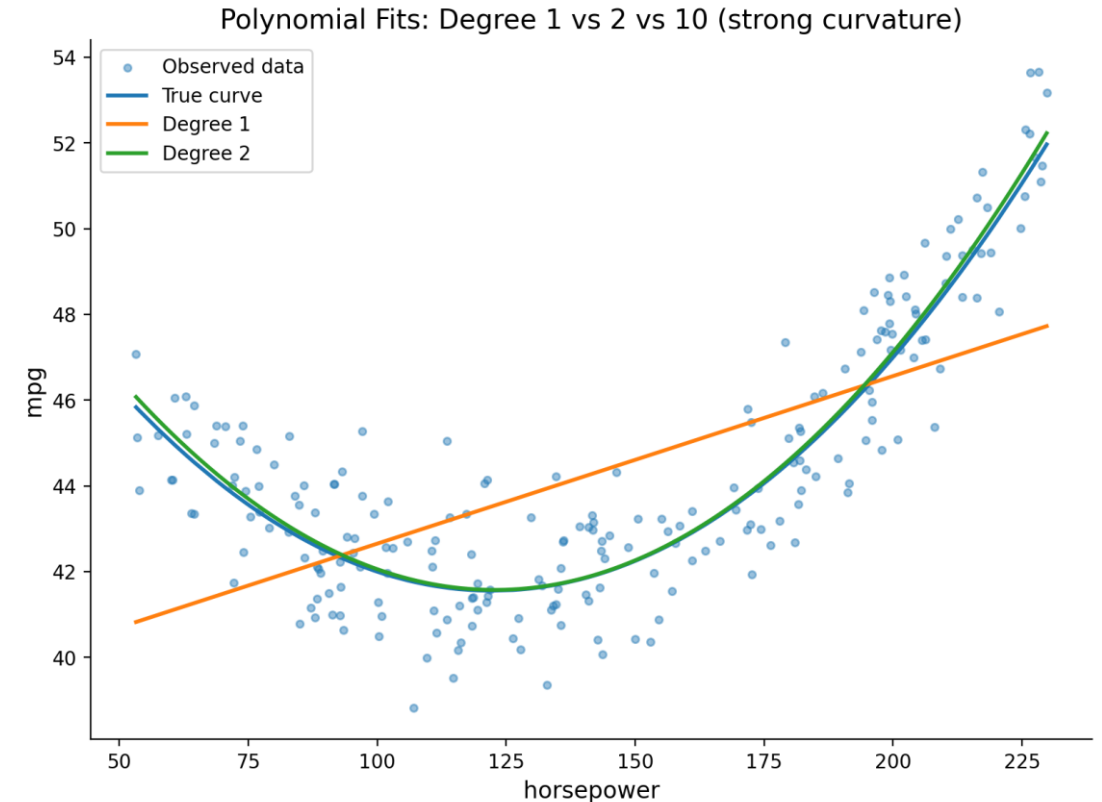
Figure: Residual plot from linear fit, showing curved U pattern

Polynomial regression can reduce residual patterns

Add squared term

$$\begin{aligned} \text{mpg} \\ = \beta_0 + \beta_1 \cdot \text{horsepower} + \beta_2 \\ \cdot \text{horsepower}^2 + \epsilon \end{aligned}$$

- Residuals become more random.
- R^2 increases (0.61 $\rightarrow$ 0.69).
- Model captures curvature without overfitting.



Polynomial regression highlights the importance of visualizing model fits

Degree 1

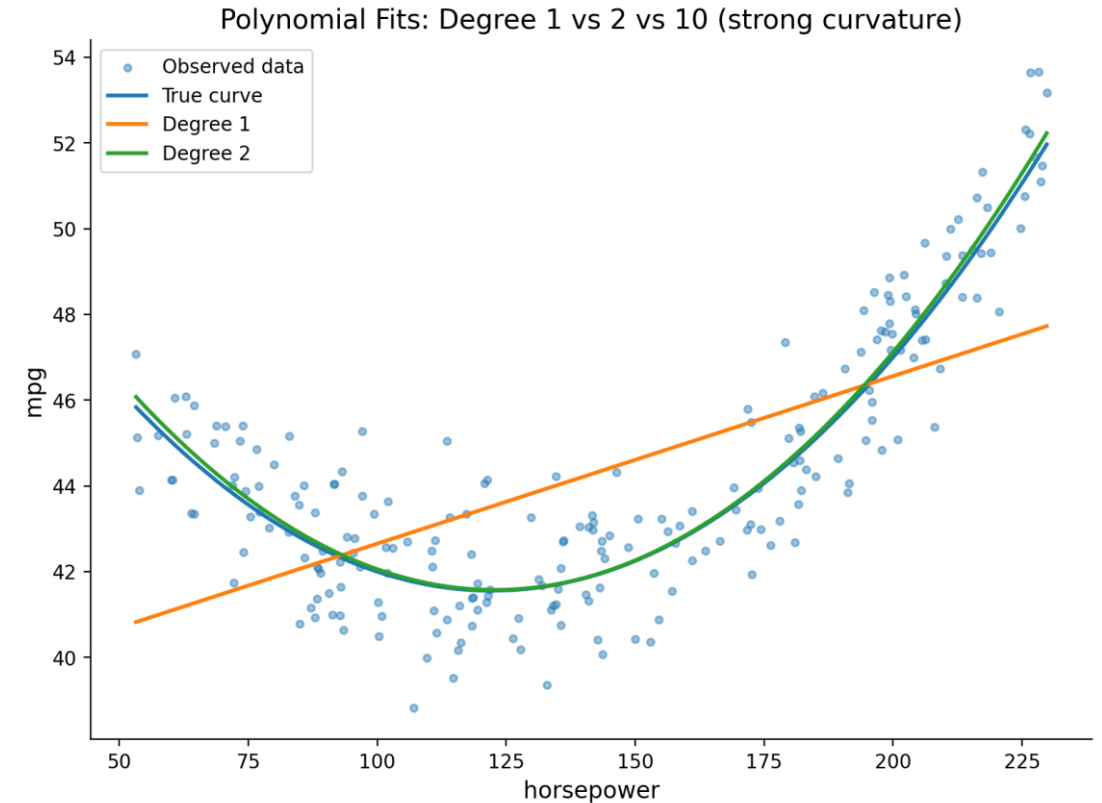
Straight line: Underfits if relationship is nonlinear

Degree 2

Parabola: Good fit for quadratic relationships

Degree 10

Highly wiggly curve: Nearly passes through every data point → overfit



Adding polynomial terms increases flexibility but risks overfitting

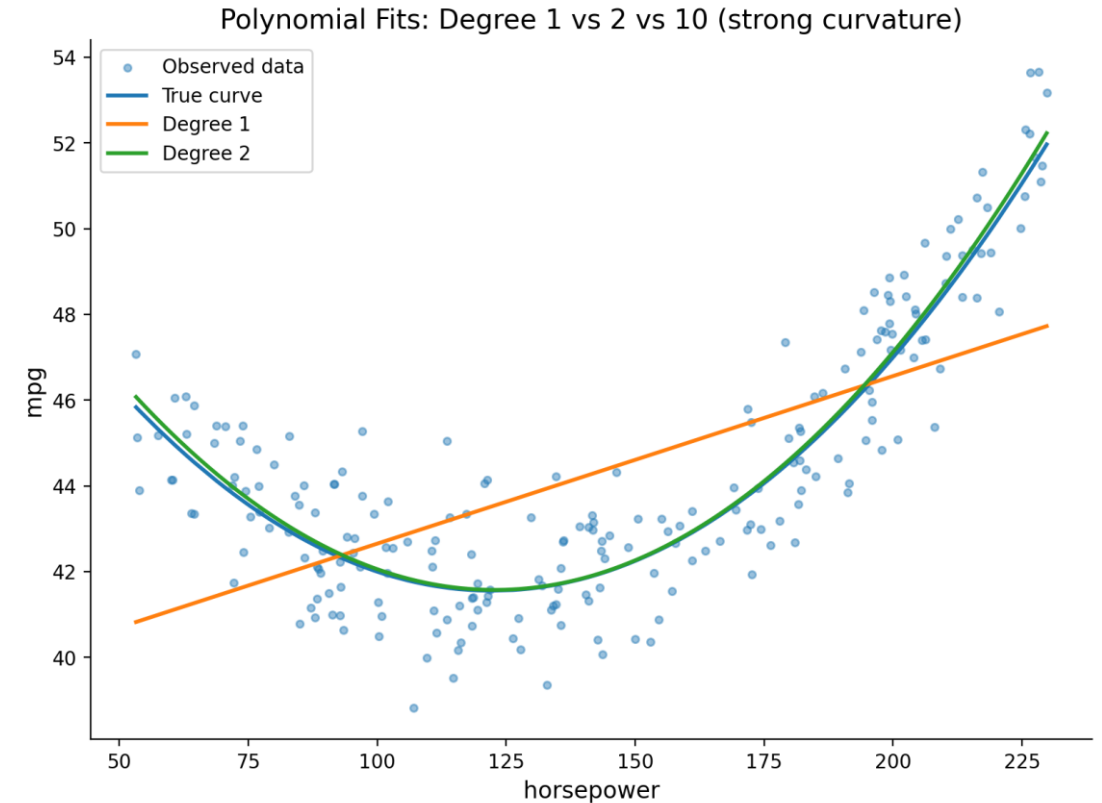
- Higher-degree polynomials allow the model to fit data more closely.

Risk

The model captures noise rather than signal.

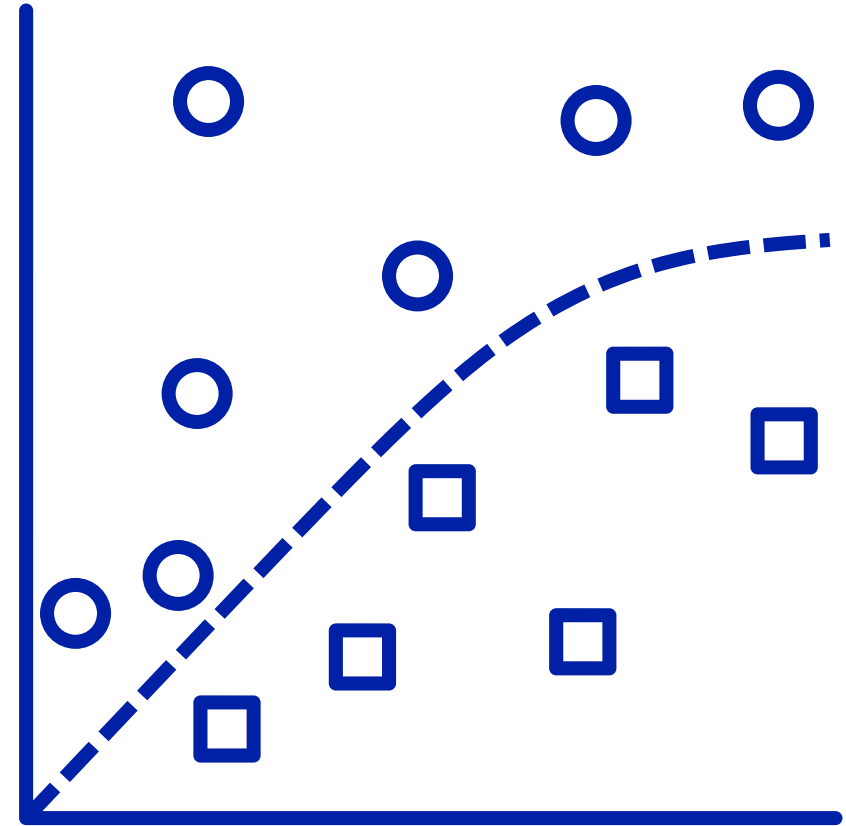
Trade-off

- Low degree:**
Underfitting: model is too simple.
 - High degree:**
Overfitting: model fits training data but fails to generalize.
- Connects directly to the **bias-variance tradeoff**.



Flexibility improves fit in-sample but can harm generalization

- Flexible models often fit the training data better.
- But too much flexibility reduces test performance.
- Polynomial regression is a classic case of **flexibility vs. generalization**.





7. Parametric vs. Nonparametric

Parametric vs. nonparametric regression

Parametric methods (e.g., OLS)

- Assume a functional form (line, polynomial).
- Estimate a small set of parameters.
- **Pros:** Interpretable, efficient with limited data

Nonparametric methods (e.g., k-nearest neighbors, trees)

- Make minimal assumptions about the form of $f(X)$.
- Fit flexible shapes directly from data.
- **Pros:** Capture complex patterns.
- **Cons:** Need more data

Linear regression vs. k-nearest neighbors

Linear regression

- Performs well if the true relationship is (approximately) linear
- Low variance, but potentially biased

KNN regression

- Predicts using the average of nearest neighbors
- Very flexible, adapts to nonlinear patterns
- High variance when (k) is small

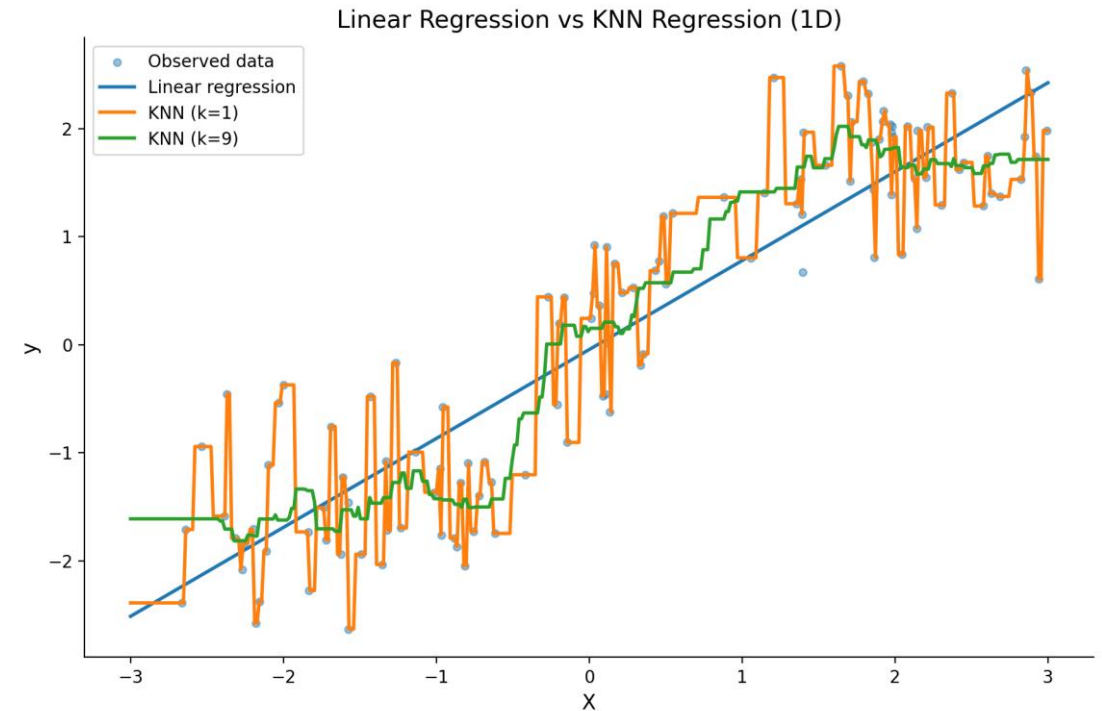


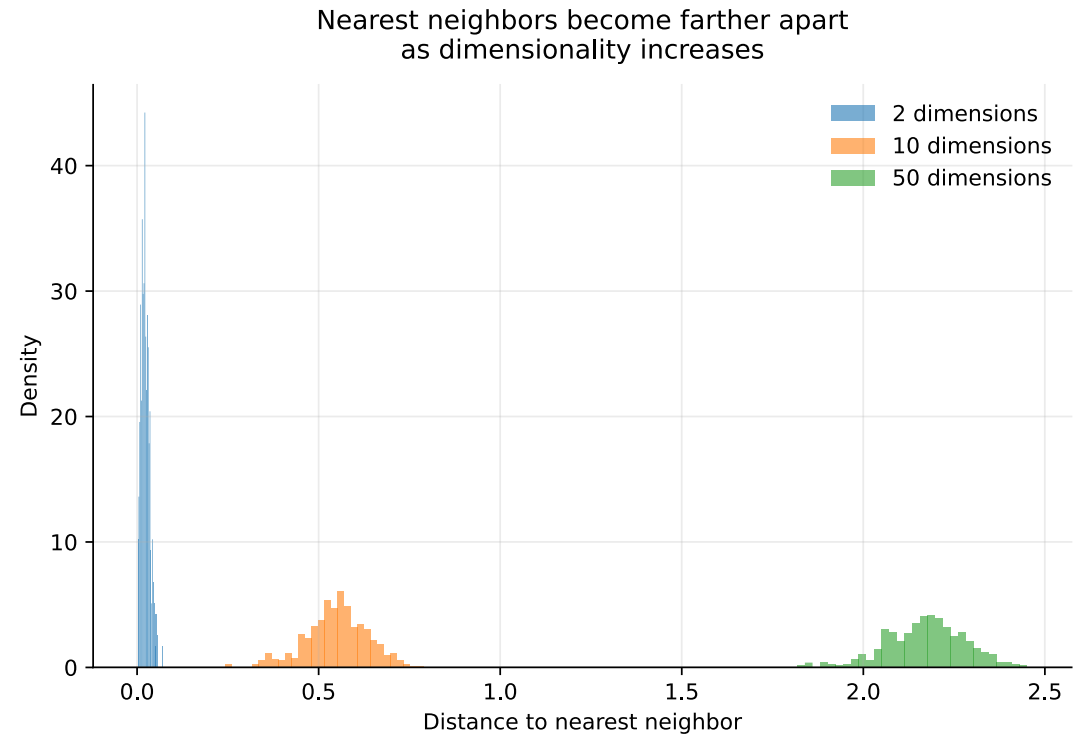
Figure: Linear regression line vs. KNN step function

The curse of dimensionality

- In high dimensions, “neighbors” are far apart.
- **KNN suffers:** Each test point has few truly close neighbors.
- Linear regression less affected by dimensionality.

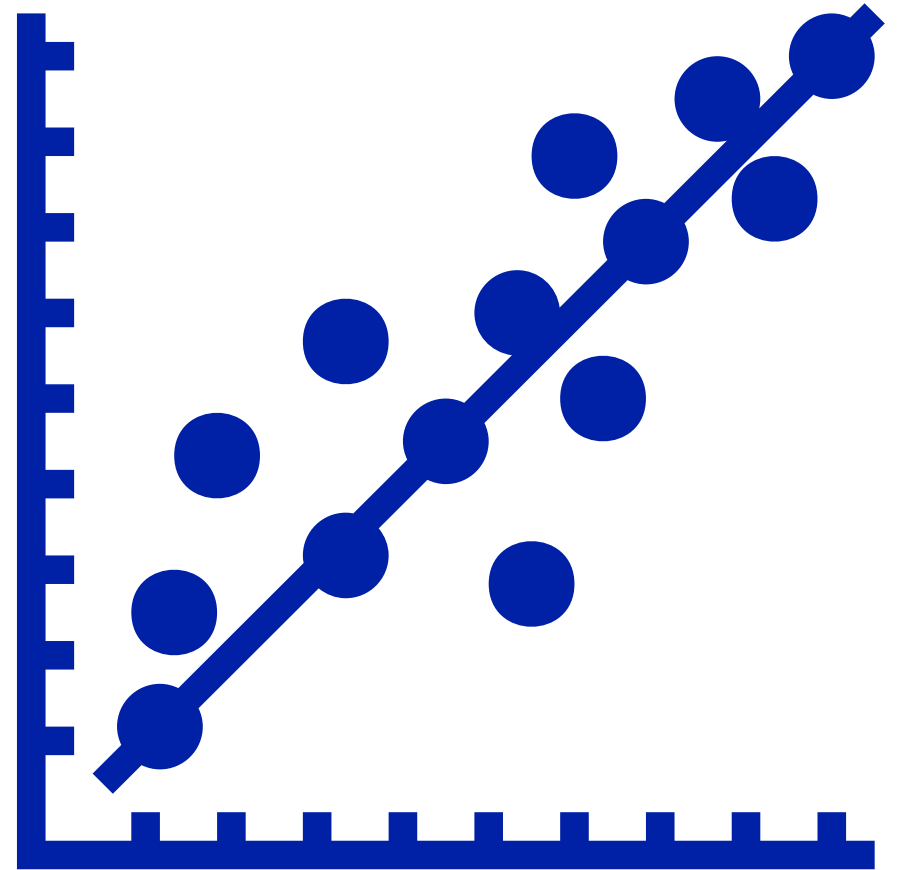
Practical lesson

- **Use LR as baseline.**
- Nonparametric methods require much larger datasets.



Why start with linear regression?

- Interpretable baseline.
- Strong performance when relationships are simple or data is limited.
- Benchmarks more complex models.
- Nonparametric methods add flexibility but need more data and care.





8. Regularization Techniques

Regularization balances bias and variance to prevent overfitting

Bias

Error from oversimplification (underfitting)

Error

Error from over-sensitivity to training data (overfitting)

Regularization

Regularization adds a penalty on large coefficients to control complexity

Visual intuition

- **High bias:** Predictions far from truth
- **High variance:** Fits noise in data
- **Regularization:** Smoother, generalizable model

Lasso regression uses an L1 penalty to shrink coefficients to zero

Loss function

$$\text{Loss} = \text{SSE} + \lambda \sum |\beta_j|$$

Effect

Some coefficients shrink to **exactly zero**.

Use

Useful for **feature selection** in high-dimensional data

Example

Predicting house prices with dozens of features
→ Lasso highlights the most important predictors

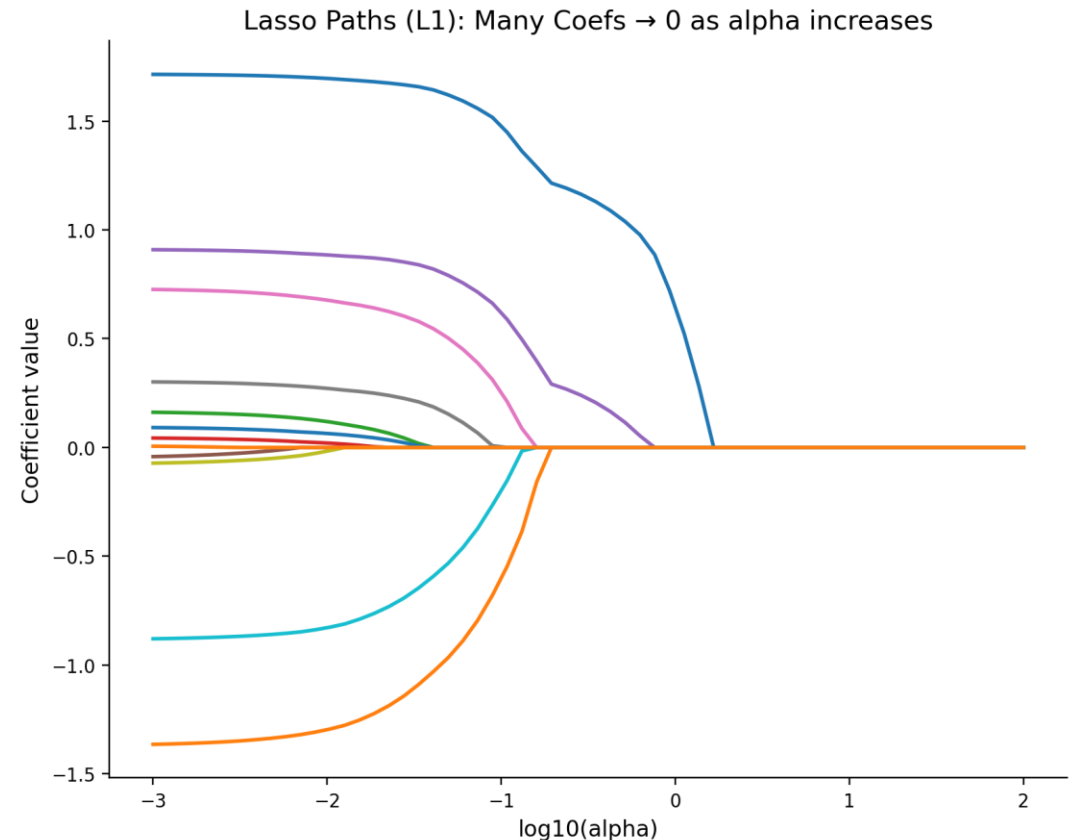


Figure: Coefficient shrinkage under Lasso regularization

Ridge regression uses an L2 penalty to shrink coefficients smoothly

Loss function

$$\text{Loss} = \text{SSE} + \lambda \sum \beta_j^2$$

Effect

Coefficients are shrunk toward zero but rarely eliminated

Use

Useful for **stabilizing models** when predictors are correlated (multicollinearity)

Example

Housing dataset with correlated features like “size in square feet” and “number of rooms”

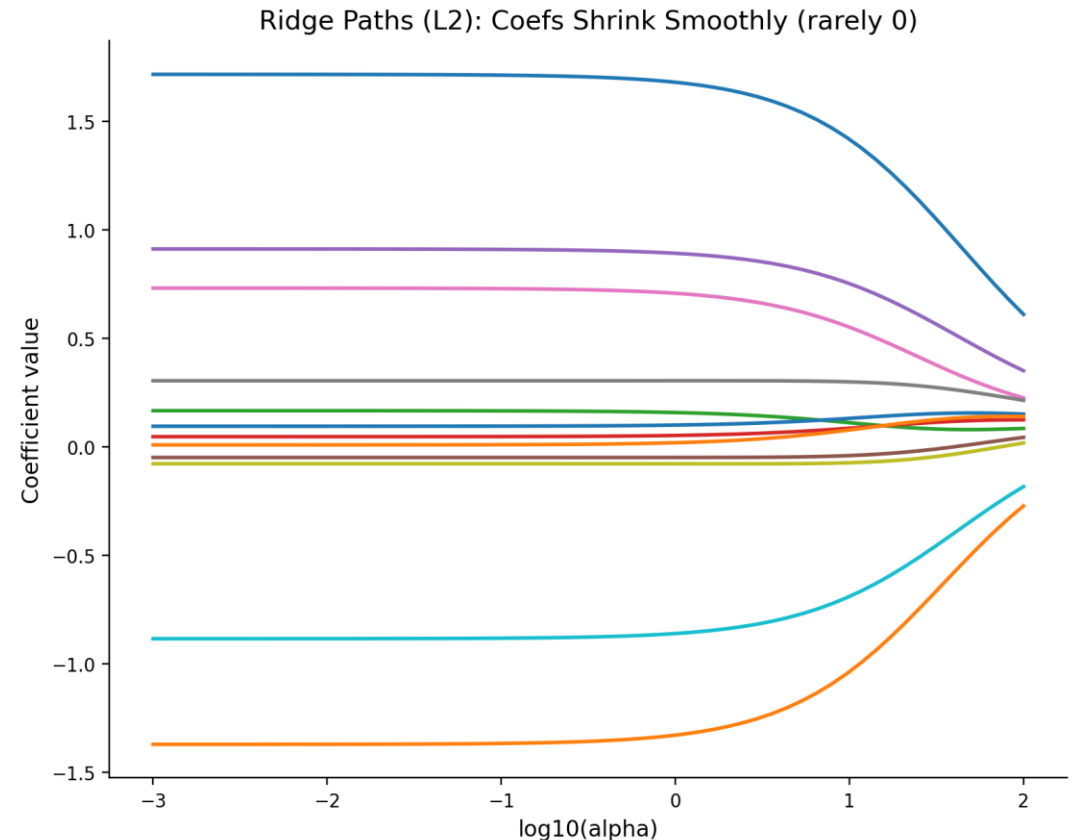


Figure: Coefficient shrinkage under Ridge regression

Elastic Net combines L1 and L2 penalties for balanced regularization

Loss function

$$\text{Loss} = \text{SSE} + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

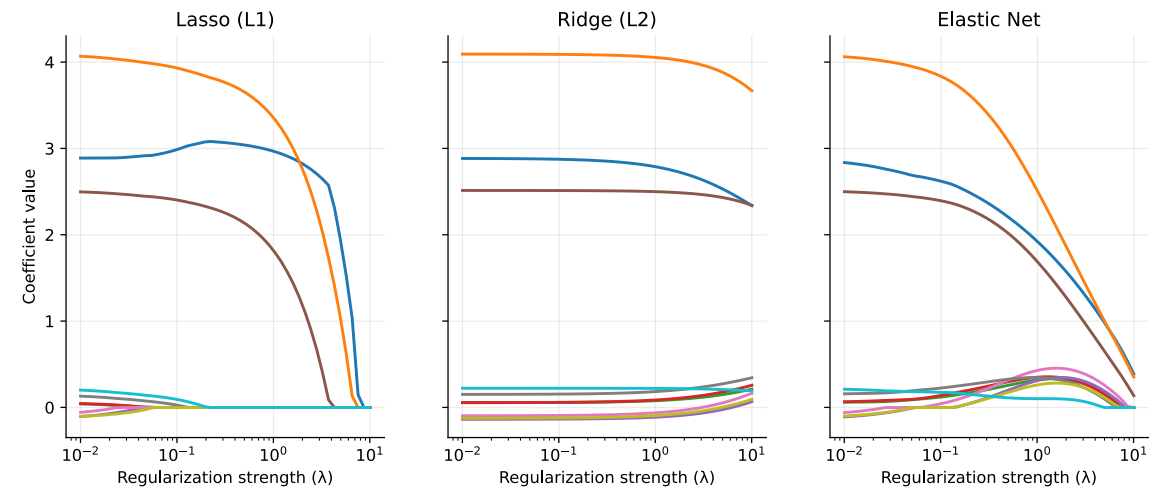
Effect

Combines sparsity (L1) with stability (L2)

Works well when:

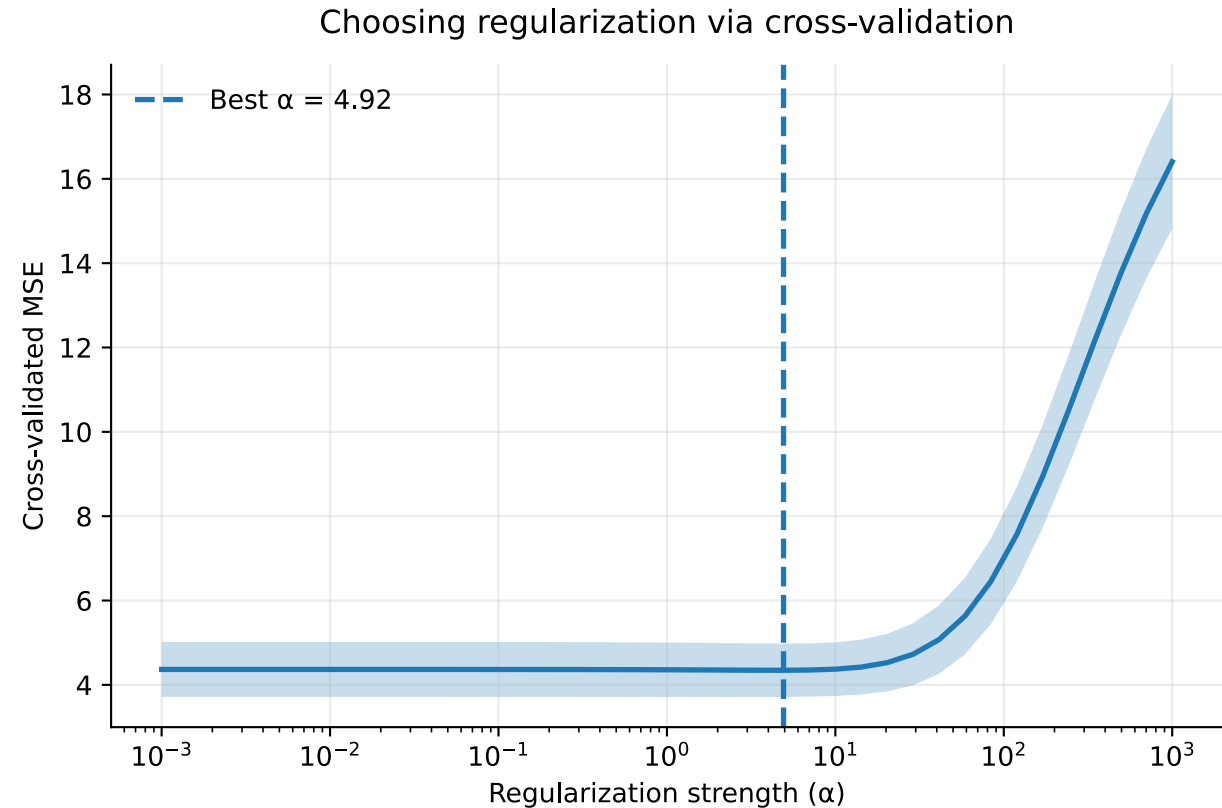
- Many correlated features exist.
- Only some predictors are expected to matter.

Elastic Net balances sparsity (L1) and stability (L2)



Hyperparameters must be tuned with cross-validation to optimize regularization

- Regularization strength (λ or α) is a hyperparameter.
- Use cross-validation to choose the best value.



High-dimensional data: When $p \geq n$

- In some settings, predictors (p) $\geq$ sample size (n).

Examples

- **Genomics:** 10,000+ genes, hundreds of samples)
- **Marketing:** Thousands of features, limited campaigns

Problem

- OLS cannot estimate unique coefficients.
- Model is not identifiable.

Regularization solves the high-dimensional problem



Penalties like Ridge, Lasso, Elastic Net

- Shrink coefficients
- Ensure unique solution even if $(p > n)$

Lasso

Selects key features, drops the rest

Ridge

Spreads weight across correlated predictors

Elastic Net

Balance of both

Key takeaway: (p n) changes the game

- **OLS fails:** Too many predictors, not enough data.
- **Regularization is required** for estimation and prediction.

Practical lesson

- Always check (p) vs. (n).
- If (p) is close to or exceeds (n), skip OLS and go straight to Ridge/Lasso/Elastic Net.

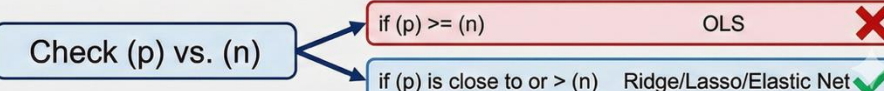
OLS Fails: Too Many Predictors (p), Not Enough Data (n)

	P1	P2	P3	P4	P5	...	...	P7	P8	...	...	P100
N1												
N2												
N3												
N4												
N7												
N8												
N10												



Regularization is Required & Practical Lesson

	P1	P2	P3	P4	P5	...	...	P6	P7	P8	...	P100
N1												
N2												
N5												
N7												
N10												





9. Evaluation Metrics for Regression

Different regression metrics capture different aspects of performance

- Metrics provide multiple perspectives on model performance.
- The best metric depends on context (goals or scientific needs).



Mean Squared Error penalizes large errors heavily

Definition

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

Properties

- **Squares errors:** Penalizes large mistakes disproportionately
- Sensitive to outliers

Example

A housing model that underestimates one mansion by \$1M gets heavily penalized

Root Mean Squared Error is interpretable in the same units as the target

Definition

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Properties

- Expressed in same units as the target
- Easier to interpret than MSE

Example

RMSE = 15 → Predictions are off by ~15 units on average. (mostly... larger errors are weighted more heavily.)

Mean Absolute Error is robust to outliers

Definition

$$\text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Properties

- Applies a linear penalty, so outliers don't dominate
- Interpretable as the average absolute deviation

Example

MAE = \$10,000 in housing → Predictions are off by \$10,000 on average.

R^2 measures the proportion of variance explained by the model

Definition

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Range

0 (no explanatory power) $\rightarrow$ 1 (perfect fit)

Caveats

- Can be negative if the model is worse than predicting the mean
- Does not always reflect predictive quality, especially in nonlinear contexts

The right regression metric depends on your context



RMSE

Common choice, interpretable, but sensitive to outliers

MAE

Robust to outliers, useful when errors should be treated equally

MSE

Good for optimization but less interpretable

R^2

Useful for communication but should not be used alone

Classical model selection approaches

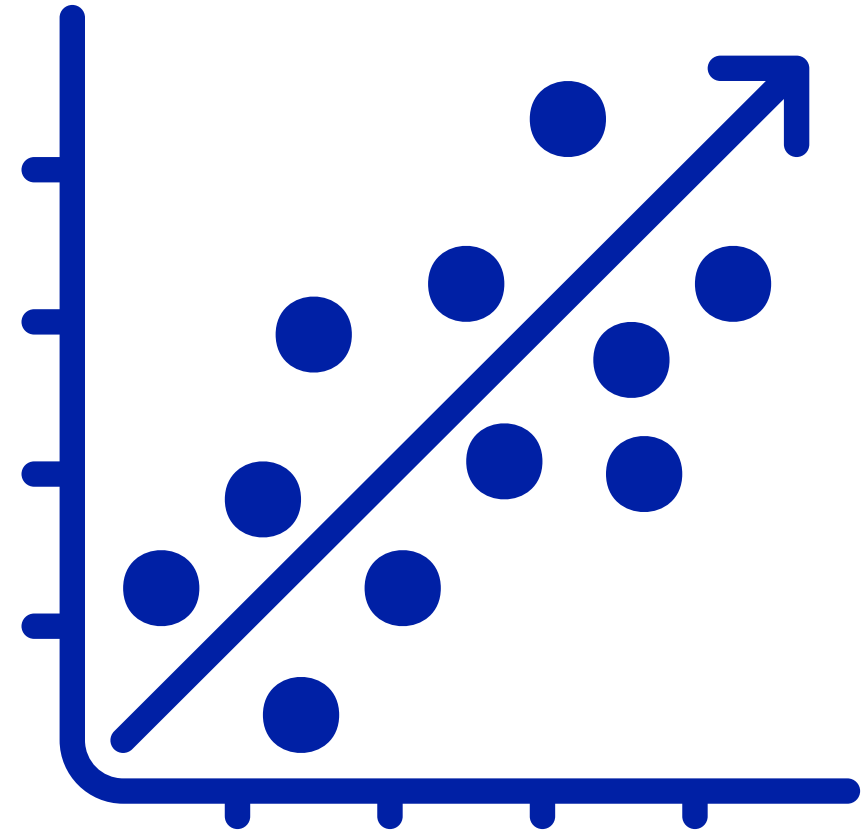
- **Early regression practice:** Compare many models with different subsets of predictors.

Tools

- **Adjusted R^2 :** Penalizes adding predictors that don't help
- **AIC / BIC:** Information criteria; lower is better

Goal

Find a model that explains data **without overfitting**



Subset selection strategies



Forward selection

Start empty → add variables one by one

Backward selection

Start with all variables → remove least useful

Stepwise / Mixed

Add and remove iteratively

Problem

- Exponential number of possible models ((2^p))
- Practical only for small (p)

Classical vs. modern model selection

Classical

- Adjusted R^2 , Cp, AIC, BIC, subset selection
- **Strength:** Interpretable, quick for small problems
- **Weakness:** Doesn't scale; prone to multiple-testing problems

Modern (preferred)

- Cross-validation for performance
- Regularization (Ridge, Lasso, Elastic Net) for complexity control

Today

Classical tools are mostly complements or quick heuristics.



10. Model Selection and Validation

Cross-validation provides more stable performance estimates



Problem

A single train/test split may give unstable results, especially with small datasets.

Solution: k-fold cross-validation

- Split data into k folds.
- Train on $k-1$ folds, validate on the remaining fold.
- Repeat k times and average results.

Benefits

- Provides a more reliable estimate of generalization.
- Reduces dependence on a single split.

The bias–variance tradeoff explains underfitting and overfitting

- **High bias (underfitting):** Model too simple → consistently wrong.
- **High variance (overfitting):** Model too complex → fits noise in training data.
- **Graph:** Test error vs. model complexity → U-shaped curve.

Examples

- **Polynomial regression:** Variance grows with degree.
- **Ridge/Lasso:** Reduce variance by shrinking coefficients.

Figure: U-shaped curves of bias–variance tradeoff

Learning curves and residual plots help diagnose model performance

Learning curves

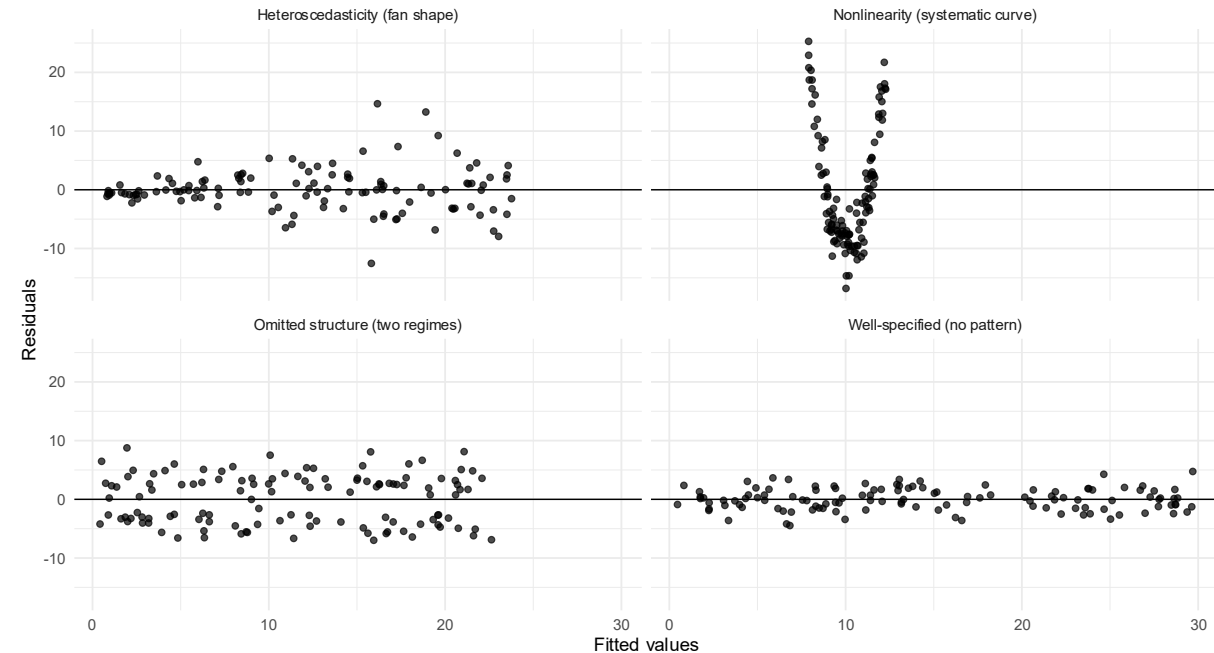
- Plot training error and validation error as the dataset size increases
- **Both high:** Underfitting
- **Large gap between them:** Overfitting
- **Both low and converged:** Good fit

Residual analysis

- **Random scatter:** Appropriate model
- **Patterns (e.g., funnel shape):** Heteroscedasticity or missing nonlinearities

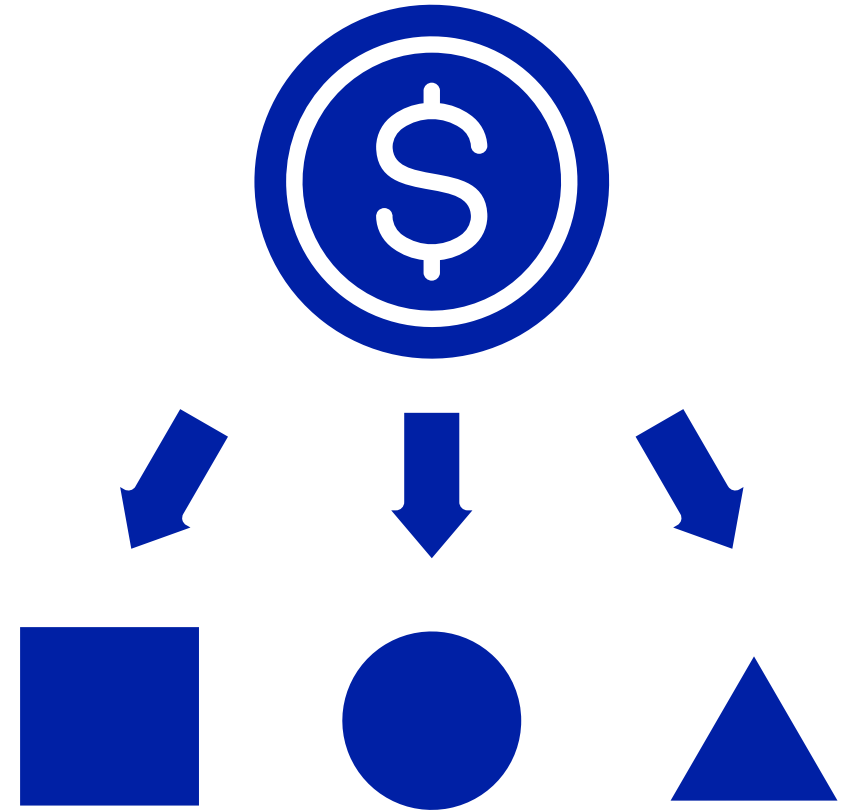
Residual patterns reveal what the model fails to capture

Systematic structure in residuals signals nonlinearity, omitted variables, or misspecification



Validation should be a habit

- Models should never be judged on training performance alone.
- Validation helps detect overfitting and guides model and feature choices.
- In industry, cross-validation is the **default baseline** for evaluating models.





Hands-On Exercise:

Apply regression concepts end-to-end